Arithmetic Expression

An arithmetic expression is a combination of operands (like number or variables) and operators (like +, -, *, /, %, etc) that are used to perform a calculation.

Eg;

A + B
5 - 8
3 * 2
A / B

→ Precedence → from left to right.

Highest → Exponential (↑)
Next Highest → Multiplication (*) and division (/)
Lowest → Addition (+) and subtraction (-).

i.e; Eg;

```
int a = 5
int b = a/2 * 4;
print"(b)"
```

output → $b = \frac{5}{2} \times 4$ (int)

$= 2 \times 4 = 8$

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Eg; $2 \uparrow 3 + 5 * 2 \uparrow 2 - 12/6$

$\downarrow$

$(2)^3 + 5 \times (2)^2 - 12/6$

$8 + 5 \times 4 - 2$

$8 + 20 - 2$

$= 26$

Now,

Most common arithmetic operation

$A + B$ $\quad C - D$ $\quad E * F$ $\quad G/H$

This is called <u>infix notation</u>, more example;

$(A+B)*C$ and $A + (B*C)$

Eg; for infix notation →

$A + B/C * (D+E) - F$

In infix notation.

more eg;

$2 + (8/4) * (5-3) + (2+3) * 5$

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{Note → In infix notation we use paranthesis '()'. (same as we use in maths)}

→ Polish Notation → Also known as prefix notation. It is a way of writing arithmetic expression without needing paranthesis '()'.

The operator (+, −, /, *, %.) comes before the operands.

Infix	Polish / Prefix
A + B	+AB
A * B	*AB
A − B	−AB
A / B	/AB
A + (B * C)	A+(*BC) = +A*BC
(A + B) * C	(+AB) * C = *+ABC
(A+B) / (C−D)	(+AB)/(−CD) = /+AB−CD

+etc;

Same as Vice-Versa

+AB = A+B

/+ABC = /(A+B)C = (A+B)/C

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Prefix into infix

Eg:

① $+\ 2\ 3 = 2+3 = 5$

② $-\ *\ 2\ 3\ 6 = -(2\times 3)\ 6$

$= -(6)\ 6$

$= 6-6 = 0$

③ $*\ /\ 8\ 2\ 3 = \left(\frac{8}{2}\right) * 3$

$= 4\times 3$

$= 12$

vice - versa

Infix into prefix

④ $(2+4)*3+(5*4) = (+\ 2\ 4 * 3) + (*\ 5\ 4)$

$= (*\ +\ 2\ 4\ 3) + (*\ 5\ 4)$

$= +\ *\ +\ 2\ 4\ 3\ *\ 5\ 4$

⑤ $2+3*(6/3-5)\uparrow(5+3)$

$= 2+3*(/\ 6\ 3-5)\uparrow(+\ 5\ 3)$

$= 2+3*(-\ /\ 6\ 3\ 5)\uparrow(+\ 5\ 3)$

$= 2+(*\ 3\ -\ /\ 6\ 3\ 5)\uparrow(+\ 5\ 3)$

$= 2+\uparrow(*\ 3\ -\ /\ 6\ 3\ 5\ +\ 5\ 3)$

$+\ 2\uparrow *\ 3\ -\ /\ 6\ 3\ 5\ +\ 5\ 3$

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⑥ : A + B/C * (D+E) - F

= A + (/BC) * (+DE) - F

= A + (* /BC + DE) - F

= (+ A * /BC + DE) - F

= (- + A * /BC + DE F)

here,

Step 1 :- pop B/C
push /BC

2 :- pop D+C
push +DC

3 :- pop /BC * +DC
push * /BC + DC

4 :- pop A + */BC +DC
push + A */BC +DC

5 :- pop +A */BC +DC - F
push - + A */BC+DC F

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→ Post-fix (reverse polish notation) :-

- Operator comes after operands.
- No paranthesis are needed.

eg; $(A+B)*C$

$= (AB+)*C$

$= AB+C*$

Infix	Post-fix
$A+B$	$AB+$
$A-B$	$AB-$
$A*B$	$AB*$
A/B	$AB/$
$A+(B*C)$	$A+(BC*) = ABC*+$
$A*(B+C)$	$A*(BC+) = ABC+*$

Same as vice-versa;

$AB+ = A+B$

$ABC+*DE-/ = A(B+C)*(D-E)/$

$= A*(B+C)(D-E)/$

$= A*(B+C)/(D-E)$

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Eg: ~~prefi~~ postfix into infix:-

① 2 5 * = 2 × 5 = 10

② 8 7 3 - / = 8 (7-3) /
= 8 4 /
= 8/4
= 2

③ 8 2 / 3 * 4 - = 8/2 3 * 4 -
= 4 × 3 4 -
= 12 - 4
= 8

vice - versa →

Infix into postfix:-

④ (2+3) * 4 + (5 - 3)

(2 3 +) * 4 + (5 3 -)

(2 3 + 4 *) + (5 3 -)

= 2 3 + 4 * 5 3 - +

⑤ (2/3) * 5 - (6 * 2) / 4

2 3 / 5 * + 6 2 * 4 / -

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given infix to postfix using stack

A+B/C*(D+E)-F

=> A+(BC/)*(DE+)-F

=> ~~ABC/+DE+*F~~

=> A+(BC/DE+*)-F

=> (ABC/DE+*+)-F

=> ABC/DE+*+F-

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Q.7.) Solve this postfix expression

8 7 3 - / 6 2 5 4 + * + -

=> => 8 (7-3) / 6 2 5 4 + * + -

=> 8/4 6 2 5 4 + * + -

=> 2 6 2 5 4 + * + -

=> 2 6 2 (5+4) * + -

=> 2 6 (2*9) + -

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$\Rightarrow 2 \quad (6+18) \quad -$

$\Rightarrow 2 - 24$

$\Rightarrow -22$

Step - wise Conversion →

Symbol	Stack
8	8
7	8,7
3	8,7,3
-	8,4
/	2
6	2,6
2	2,6,2
5	2,6,2,5
4	2,6,2,5,4
+	2,6,2,9
*	2,6,18
+	2,24
-	-22

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Q6 → step wise

Symbol	Expression
A	A
+	A
B	AB
/	AB
C	ABC/
*	ABC/
(	ABC/
D	ABC/D
+	ABC/D
E	ABC/DE
)	ABC/DE+*
-	ABC/DE+*+
F	ABC/DE+*+F
)	ABC/DE+*+F-

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Q.8. Evaluate the postfix

8 2 / 3 * 4 - 6 2 / +

⇒ 8/2 3*4-6 2/+

4*3 4-6 2/+

12-4 6 2/+

8 (6/2)+

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8 + 3
= 11 //.

Step-wise

Symbol	Stack
8	8
2	8, 2
/	4
3	4, 3
*	12
4	12, 4
-	8
6	8, 6
2	8, 6, 2
/	8, 3
+	11 //.

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